



Simple 5W Mono Audio Amplifier

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This mono AF amplifier uses a single 5-pin LA4425A IC to deliver 5-watt output through a 4-ohm, 5-watt loudspeaker. The

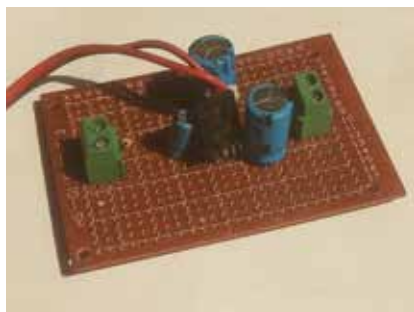


Fig. 1: Author's prototype

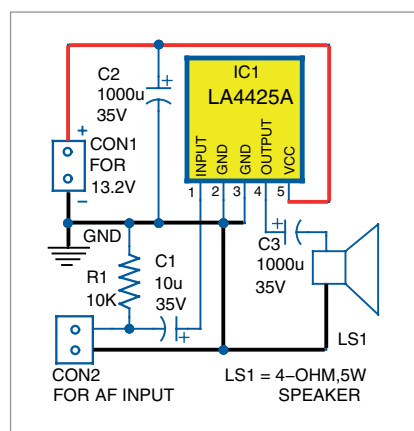


Fig. 2: Circuit diagram of the amplifier

PARTS LIST

Semiconductors:

IC1 - LA4425A audio power amplifier

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1 - 10-kilo-ohm

Capacitors:

C1 - 10µF, 35V electrolytic

C2, C3 - 1000µF, 35V electrolytic

Miscellaneous:

CON1, CON2 - 2-pin connector

LS1 - 4-ohm, 5W speaker

- Heat-sink for IC1

- 12-13.2V DC power supply

- 2-pin connector for LS1

datasheet of LA4425A recommends a 13.2V power supply to get 5-watt output. A suitable heat-sink may be used for the IC for protection, especially if a higher voltage supply is used. The author's prototype is shown in Fig. 1.

Circuit and working

Circuit diagram of the simple amplifier, which is built around audio amplifier LA4425A (IC1), is shown in Fig. 2.

Pin 1 of LA4425A is connected to AF input terminal through electrolytic capacitor C1. An optional resistor (R1) is connected to ground at the input. Pins 2 and 3 of IC1 are tied together and connected to ground terminal. Power supply is connected to pin 5 of IC1. Loudspeaker (LS1) is connected to output pin 4 of IC1 through electrolytic capacitor C3.

Construction and testing

An actual-size PCB layout for the amplifier is shown in Fig. 3 and its components layout in Fig. 4. Even a 5cm $\times$ 7cm Veroboard can be used in place of the PCB for construction of this amplifier.

Mount 2-pin connectors on the left and right side of the PCB (or Veroboard) for power supply and loudspeaker, respectively. The IC may be placed in the middle of the board and the remaining components mounted as per circuit diagram. You may use a small aluminium sheet as heat-sink for the IC.

Connect the speaker to LS1 terminals on the board and a 12V battery

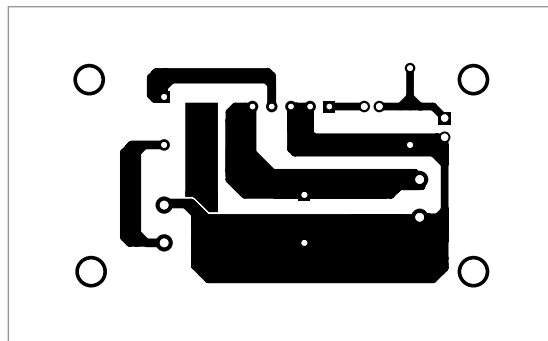


Fig. 3: Actual-size PCB layout

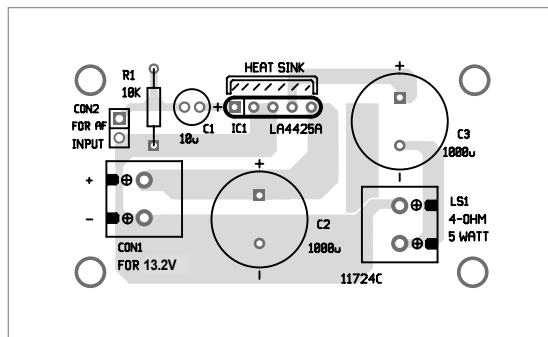


Fig. 4: Components layout

to the power supply terminals CON1. You may connect a 10-kilo-ohm potentiometer (pot) as volume control at input of the amplifier.

To test the amplifier, take a small metal screwdriver and gently touch at input of the amplifier. If you hear a humming sound from the speaker, your amplifier is ready to use.

You may now connect an audio source, such as MP3 player or laptop, to input connector CON2. Use the pot to adjust volume of the sound from the speaker and enjoy. **EFY**



Raj K. Gorkhali is an electronics hobbyist and a regular contributor to EFY